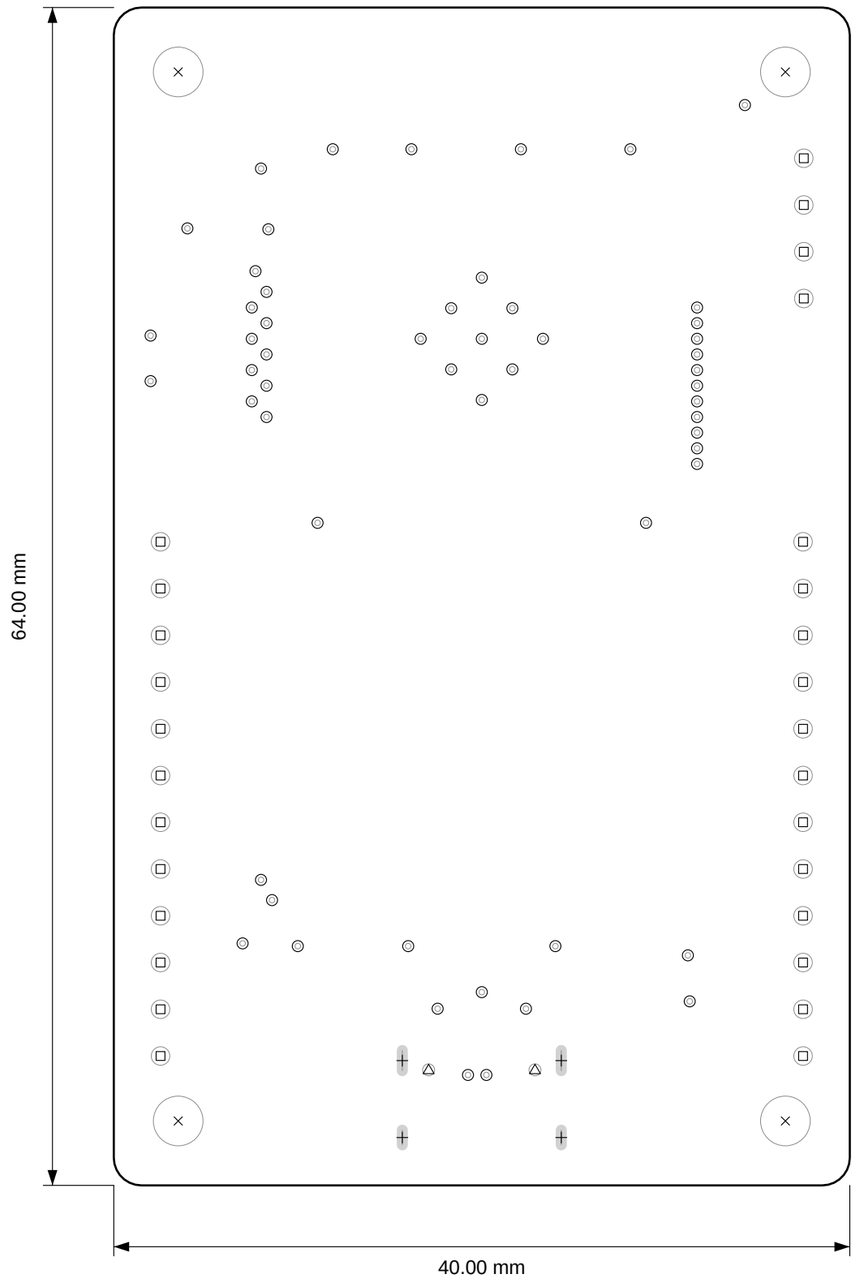
ESP32-S31 dev board — fabrication drawing

Viewed from the top. Scale 3:1. All dimensions in mm.



Drill table (93 holes, finished sizes)

Sym	Tool	Ø mm	Plated	Kind	Use	Count
◯	T1	0.300	yes	hole	via	55
+	T2	0.600	yes	slot	component	4
□	T3	1.005	yes	hole	component	28
△	T4	0.650	no	hole	mechanical	2
×	T5	2.700	no	hole	mechanical	4

Stackup (4 copper layers, 1.74 mm over copper)

Layer	Material	mm
Top legend	white ink	—
Top mask	green solder mask	0.012
Top copper	35 µm Cu	0.035
Prepreg 1	FR-4 2116	0.200
Inner 1 copper	35 µm Cu	0.035
Core 1	FR-4 7628	1.200
Inner 2 copper	35 µm Cu	0.035
Prepreg 2	FR-4 2116	0.200
Bottom copper	35 µm Cu	0.035
Bottom mask	green solder mask	0.012
Bottom legend	white ink	—

Notes

1. Nominal board thickness 1.74 over the outer copper, solder mask and legend excluded; 4 copper layers.
2. Solder mask 12 µm thick over copper on each masked side.
3. Copper thickness top to bottom: 35 µm, 35 µm, 35 µm, 35 µm.
4. Minimum track width 0.09, minimum copper gap 0.09.
5. Minimum finished drill 0.15, hole to copper 0.2.
6. Minimum annular ring: vias 0.05, plated holes 0.15.
7. Solder mask: green gloss, expansion 0, minimum web 0.1.
8. Legend: white ink, minimum line 0.15, minimum text height 1.
9. Surface finish: ENIG.
10. Dimensions in millimetres. Hole sizes are finished sizes; the drill file lists finished sizes. The fabricator does not publish how it reads drill files: confirm with it.
11. Generated by Pcbcraft; the Gerber job file lists every layer.

ESP32-S31 dev board

Drawing	Fabrication drawing, sheet 1 of 1
Rules	JLCPCB 4-layer 1 oz (from JLCPCB, PCB Manufacturing & Assembly Capabiliti…
Board	40.00 × 64.00 mm, 4 layers, 1.74 mm
Finish	ENIG
Mask / legend	green / white
Generator	Pcbcraft